

Postprandial plasma lipid hydroperoxides: a possible link between diet and atherosclerosis.

There is increasing evidence implicating a dietary source of plasma lipid peroxides that become elevated in the postprandial state. This phenomenon may be a contributing factor to the correlation found between postprandial hyperlipidemia and increased risk of cardiovascular disease. Using a newly developed method for measuring lipid hydroperoxides directly in plasma, a pilot study was performed which revealed that lipid hydroperoxides are indeed elevated following a fatty meal. Lipid hydroperoxides increased within 2-4 h after the meal and returned to basal levels, corresponding to the usual postprandial hyperlipidemia. A marked suppression of postprandial hydroperoxides was found when a meal was consumed with wine, suggesting that these hydroperoxides can be formed and then absorbed during the digestive process.